

BNPL Platform Performance & Transaction Approval Enhancement

Business Requirements Document (BRD)

Project Case Study

Document Information	Details
Project	BNPL Platform Performance & Transaction Approval Enhancement
Business Area	Buy Now, Pay Later (BNPL) - Transaction Approval
Document Type	Business Requirements Document (BRD)
Version	1.0
Status	Working BRD - requirements substantially validated; identified open items retained
Purpose	Define the business need, scope, stakeholders, current-state issues, target-state process, and requirements for improving BNPL transaction approval performance.

1. Executive Overview

The BNPL platform has experienced deterioration in transaction approval performance and processing time, accompanied by increased customer and merchant complaints. Analysis identified a material increase in technical declines and processing delays, particularly in the Mobile App channel. Root-cause analysis linked the issue to an incomplete change impact assessment that failed to identify an existing Mobile App identity-verification dependency, allowing duplicate verification requests in some transaction flows.

2. Business Need

Improve the reliability and performance of the BNPL transaction approval process to support customer experience, merchant satisfaction, transaction conversion, and GMV growth while maintaining required risk controls.

3. Problem Statement

Over the past three months, the BNPL platform has experienced an increase in customer and merchant complaints related to declined and delayed transactions. Stakeholders reported that the transaction approval rate decreased from a historical average of 82% to approximately 71%, below the 85% target, while average transaction processing time increased from approximately 4 seconds to 7 seconds, above the target of 5 seconds. Merchant inquiries regarding transaction status also increased, negatively affecting customer experience, merchant satisfaction, transaction conversion, and GMV growth.

Data-validation note: Project transaction data produced a pre-issue approval rate of 76.4% (Mar-May) and an issue-period rate of 66.9% (Jun-Aug). These analytical findings are retained separately from the stakeholder-reported historical baseline rather than silently replacing it.

4. Business Objectives & Success Metrics

ID	Objective	Success Measure
OBJ-01	Increase transaction approval performance while maintaining applicable risk controls.	Approval Rate $\geq$ 85%
OBJ-02	Reduce transaction processing time.	Average transaction processing time $\leq$ 5 seconds
OBJ-03	Reduce customer and merchant complaints associated with delayed transactions and technical declines.	Complaint baseline/target to be established when validated complaint data is available.

5. Project Scope

5.1 In Scope

- Improve the BNPL transaction approval workflow.
- Eliminate duplicate identity-verification requests for the same transaction within a single approval flow.
- Improve identity-verification performance visibility and monitoring.
- Define exception handling for identity-verification timeout or service unavailability.
- Conduct change impact assessment for affected channels, integrations, dependencies, and teams.
- Perform end-to-end regression testing across affected channels before production deployment.

5.2 Out of Scope

- Mobile App user-interface redesign or visual changes unrelated to the transaction issue.
- Merchant settlement process changes.
- Changes to core risk policies or risk appetite solely to increase approval rate, unless separately reviewed and approved by the Risk Team.

6. Stakeholders

Stakeholder	Power / Interest	Role in Project	Engagement
Head of BNPL Operations	High / High	Provide operational requirements, KPIs, pain points, expected outcomes, and validate operational changes.	Regular meetings, KPI reviews, key decision reviews
Product Owner	High / High	Define priorities, clarify requirements, prioritize backlog, and validate business value.	Requirements workshops, backlog reviews
Risk Team	High / High	Define and validate risk requirements/rules and ensure the solution maintains risk controls.	Risk-rule reviews, workshops, solution validation
Data Team	Low / High	Provide transaction data, definitions, data-quality validation, and analytical support.	Data requests, validation sessions
Engineering	Low / High (delivery team)	Assess feasibility, investigate dependencies, provide logs/technical evidence, and implement approved changes.	Technical workshops, requirement reviews
QA Team	Low / High (delivery team)	Validate requirements, regression/integration behavior, defects, and release readiness.	Test-case reviews, defect meetings, testing coordination

7. Current State (AS-IS) Summary

The customer initiates checkout, selects BNPL and a payment plan, and confirms the transaction. The Mobile App captures transaction/customer information and submits the transaction to the BNPL Platform. The platform validates the information and evaluates risk criteria. Where identity verification is required, the affected Mobile App flow may trigger an existing checkout verification request while the new risk workflow also triggers a verification request for the same transaction. The external identity-verification service processes the request(s), after which decisioning continues and the transaction is returned as Approved, Declined, or Pending.

AS-IS issue: The analysis established duplicate verification requests in some Mobile App flows, but did not establish an exact temporal order between the two calls. The process model therefore should not invent a sequence that was not evidenced.

8. Data Analysis Findings

Finding	Pre-Issue (Mar-May)	Issue Period (Jun-Aug)	Change / Interpretation
Overall Approval Rate	76.4%	66.9%	-9.5 percentage points
Mobile App Approval Rate	74.7%	60.2%	-14.5 percentage points
Mobile App Technical-Error Declines	58	203	+250% approximately
Mobile App Avg. Processing Time	3.81 sec	6.99 sec	+83.5%; issue period above <=5 sec target

Processing-time deterioration was also observed across other channels, indicating a potential platform-wide performance issue with a more pronounced Mobile App impact.

9. Root Cause Analysis

A platform update deployed near the end of May introduced additional real-time risk checks. Engineering evidence indicated that decisioning-service response time increased from approximately 1.4 to 2.1 seconds, while the external identity-

verification integration increased from approximately 0.8 to 2.6 seconds. Timeouts/errors were particularly elevated in the Mobile App.

Root cause: An incomplete change impact assessment failed to identify the existing Mobile App identity-verification dependency. As a result, the new risk-check workflow introduced an additional verification call, creating duplicate requests in some Mobile App transactions. The issue was not detected before release because the affected Mobile App flow was not included in the end-to-end regression testing scope.

10. Gap Analysis

Gap ID	AS-IS	Required / TO-BE State	Related Requirement
GAP-01	Multiple identity-verification requests may be triggered for the same Mobile App transaction.	No more than one verification request per transaction during a single approval flow.	FR-01
GAP-02	No clearly defined exception handling for verification timeout/service unavailability.	Defined retry and Pending behavior for technical service failure.	FR-02
GAP-03	Insufficient visibility into verification response time, failures, and timeouts.	Monitoring capability for identity-verification performance and failures.	FR-03
GAP-04	Change impact assessment/testing did not sufficiently identify channel dependencies or cover the affected end-to-end flow.	Impact assessment before implementation and E2E regression across affected channels before production.	TR-01, TR-02

11. Target State (TO-BE) Summary

The target transaction flow retains validation and risk evaluation but allows no more than one identity-verification request for the same transaction during a single approval flow. If verification is required, the platform triggers one request, receives the verification result, and continues decisioning. If the verification service times out or is unavailable, the system retries once; if the retry is unsuccessful, the transaction is set to Pending and the customer is informed that the transaction is temporarily under processing. Monitoring is introduced for identity-verification response times, failures, and timeouts.

A separate target change/release process requires change impact assessment to identify affected channels, integrations, dependencies, and teams, followed by end-to-end regression testing before production deployment.

12. Requirements Register

ID	Type	Requirement	Priority	Status
BR-01	Business	The BNPL platform shall achieve a transaction approval rate of at least 85% while maintaining an average transaction processing time of no more than 5 seconds.	Must	Validated
SR-01	Stakeholder	The Risk Team requires the effectiveness of existing risk controls to be maintained while improving transaction approval and processing performance.	Must	Validated
BRULE-01	Business Rule	If a transaction meets the defined risk criteria,	Must	Validated

ID	Type	Requirement	Priority	Status
		the system shall trigger identity verification.		
FR-01	Functional	The system shall trigger no more than one identity-verification request for the same transaction during a single approval flow.	Must	Validated
FR-02	Functional - Exception Handling	If identity verification is required and the verification service times out or is unavailable, the system shall automatically retry the verification request once. If the retry is unsuccessful, the transaction shall be set to Pending and the customer shall be informed that the transaction is temporarily under processing.	Must	Validated in project scenario
FR-03	Functional - Monitoring	The system shall provide monitoring of identity-verification response times, failures, and timeouts.	Must	Needs detailed alert criteria
NFR-01	Non-Functional - Performance	The identity-verification service shall respond within 2 seconds for at least 95% of requests.	Should	Validated
TR-01	Transition	Before production deployment, the updated BNPL approval workflow shall undergo end-to-end regression testing across all affected channels.	Must	Validated
TR-02	Transition / Process	Before implementing changes to the BNPL approval workflow, affected channels, integrations, and system dependencies shall be identified and documented through a change impact assessment.	Must	Classification to finalize

13. Assumptions & Constraints

13.1 Assumptions

- The existing external Identity Verification provider will continue to be used.
- The Data Team will provide accurate transaction data for analysis and validation.

13.2 Constraints

- The solution must comply with existing Risk policies.

- The project must improve performance without removing required identity-verification controls.

14. Data Quality & Validation Summary

DQ ID	Issue	Correction	Status
DQ-001	Missing Decline Reason (TX002756)	Blank -> Technical Error	Completed
DQ-002	Negative Approval Time (TX005211)	-2.5 -> 2.5 sec	Completed
DQ-003	Negative Transaction Amount (TX001045)	-350 -> 350	Completed
DQ-004	Duplicate Transaction ID	TX000377 -> TX006001 for one legitimate extracted record	Completed
DQ-005	Duplicate Transaction ID	TX001901 -> TX006002 for one legitimate extracted record	Completed
DQ-006	Duplicate Transaction ID	TX004112 -> TX006003 for one legitimate extracted record	Completed

Post-cleaning validation found no remaining critical completeness, duplicate, validity, or consistency issues in the project dataset.

15. Open Items / Pending Decisions

Item	Required Decision / Detail	Owner(s)
FR-03 Monitoring	Define alert thresholds, alert recipients, monitoring channel/dashboard, and notification frequency.	Platform / Operations / Product Owner / QA
TR-02 Classification	Confirm whether Change Impact Assessment remains a project-specific transition requirement or becomes an ongoing process/business rule.	Product Owner / Governance / QA
Complaint KPI	Establish validated complaint baseline and target to make Objective 3 quantitatively measurable.	Operations / Data Team
Final Sign-off	Formal business, risk, product, engineering, and QA approval after outstanding details are resolved.	Key stakeholders

16. Acceptance & Sign-off

The BRD is ready for formal sign-off when all open items are resolved, requirements are agreed by accountable stakeholders, and the target-state process is confirmed. Detailed Agile acceptance criteria, backlog items, RTM, UAT scenarios, and release validation will be maintained in subsequent project artifacts.

Role	Name	Decision	Date
Head of BNPL Operations			
Product Owner			
Risk Representative			
Engineering Representative			
QA Representative			